

Empirical Bayes and Shrinkage Estimators

Many of the classical ideas in statistics become less reliable when there are many parameters. Results in this chapter suggest a natural approach in some situations and illustrate one way in which classical ideas may fail.

11.1 Empirical Bayes Estimation

Suppose several objects are measured using some device and that the measurement errors are i.i.d. from $N(0, 1)$. (Results here can easily be extended to the case where the errors are from $N(0, \sigma^2)$ with σ^2 a known constant.) If we measure p objects then our data $X_1, \dots, X_p$ are independent with

$$X_i \sim N(\theta_i, 1), \quad i = 1, \dots, p,$$

where $\theta_1, \dots, \theta_p$ are the unknown true values. Let X denote the vector $(X_1, \dots, X_p)'$ and θ the vector $(\theta_1, \dots, \theta_p)'$. If we estimate θ_i by $\delta_i(X)$ and incur squared error loss, then our total loss, called *compound squared error loss*, is

$$L(\theta, \delta) = \sum_{i=1}^p (\theta_i - \delta_i(X))^2.$$

Note that the framework here allows the estimator $\delta_i(X)$ of θ_i to depend on X_j for $j \neq i$. This is deliberate, and although it may seem unnecessary or unnatural, some estimators for θ_i in this section depend on X_i and to some extent the other observations. This may be an interesting enigma to ponder as you read this section. Letting $\delta(X)$ denote the vector $(\delta_1(X), \dots, \delta_p(X))'$, the compound loss $L(\theta, \delta)$ equals $\|\delta(X) - \theta\|^2$, and the risk function for δ is given by

$$R(\theta, \delta) = E_\theta \|\delta(X) - \theta\|^2.$$

At this stage, let us consider a Bayesian formulation in which the unknown parameter is taken to be a random variable Θ . For a prior distribution, let

$\Theta_1, \dots, \Theta_p$ be i.i.d. from $N(0, \tau^2)$. Given $\Theta = \theta$, $X_1, \dots, X_p$ are independent with $X_i \sim N(\theta_i, 1)$, $i = 1, \dots, p$. Then the conditional density of X given $\Theta = \theta$ is

$$\frac{1}{(2\pi)^{p/2}} \exp\left[-\frac{1}{2} \sum_{i=1}^p (x_i - \theta_i)^2\right],$$

the marginal density of Θ is

$$\frac{1}{(2\pi\tau^2)^{p/2}} \exp\left[-\frac{1}{2\tau^2} \sum_{i=1}^p \theta_i^2\right],$$

and, multiplying these together, the joint density of X and Θ is

$$\frac{1}{(2\pi\tau)^p} \exp\left[-\frac{1}{2} \sum_{i=1}^p (x_i - \theta_i)^2 - \frac{1}{2\tau^2} \sum_{i=1}^p \theta_i^2\right].$$

Completing the square,

$$(x_i - \theta_i)^2 + \frac{\theta_i^2}{\tau^2} = \left(1 + \frac{1}{\tau^2}\right) \left(\theta_i - \frac{x_i}{1 + 1/\tau^2}\right)^2 + \frac{x_i^2}{1 + \tau^2}.$$

Integrating against θ , the marginal density of X is

$$\begin{aligned} \int \dots \int \frac{\exp\left[-\frac{1}{2} \left(1 + \frac{1}{\tau^2}\right) \sum_{i=1}^p \left(\theta_i - \frac{x_i}{1 + 1/\tau^2}\right)^2 - \frac{1}{2} \sum_{i=1}^p \frac{x_i^2}{1 + \tau^2}\right]}{(2\pi\tau)^p} d\theta \\ = \frac{1}{(2\pi(1 + \tau^2))^{p/2}} \exp\left[-\frac{1}{2} \sum_{i=1}^p \frac{x_i^2}{1 + \tau^2}\right]. \end{aligned}$$

This is a product of densities for $N(0, 1 + \tau^2)$, and so $X_1, \dots, X_p$ are i.i.d. from $N(0, 1 + \tau^2)$. Dividing the joint density of X and Θ in the Bayesian model by the marginal density of X , the conditional density of Θ given $X = x$ is

$$\frac{1}{\left(2\pi \frac{\tau^2}{1 + \tau^2}\right)^{p/2}} \exp\left[-\frac{1}{2} \sum_{i=1}^p \frac{\left(\theta_i - \frac{x_i}{1 + 1/\tau^2}\right)^2}{\tau^2/(1 + \tau^2)}\right].$$

Noting that this factors into a product of normal densities, we see that given $X = x$, $\Theta_1, \dots, \Theta_p$ are independent with

$$\Theta_i | X = x \sim N\left(\frac{x_i}{1 + 1/\tau^2}, \frac{\tau^2}{1 + \tau^2}\right).$$

From this

$$E[\Theta_i|X] = \frac{X_i}{1 + 1/\tau^2} \text{ and } \text{Var}(\Theta_i|X) = \frac{\tau^2}{1 + \tau^2}.$$

The expected loss under the Bayesian model is

$$\begin{aligned} EL(\Theta, \delta(X)) &= E \sum_{i=1}^p (\delta_i(X) - \Theta_i)^2 \\ &= EE \left[\sum_{i=1}^p (\delta_i(X) - \Theta_i)^2 \mid X \right] \\ &= E \sum_{i=1}^p E \left[(\delta_i(X) - \Theta_i)^2 \mid X \right] \\ &= E \sum_{i=1}^p \left[\text{Var}(\Theta_i|X) + \left(\frac{X_i}{1 + 1/\tau^2} - \delta_i(X) \right)^2 \right]. \end{aligned}$$

This risk is minimized taking

$$\delta_i(X) = \frac{X_i}{1 + 1/\tau^2} = \left(1 - \frac{1}{1 + \tau^2} \right) X_i. \quad (11.1)$$

In the Bayesian approach to this problem here, the choice of τ is crucial. In an *empirical Bayes* approach to estimation, the data are used to estimate parameters of the prior distribution. To do this in the current setting, recall that under the Bayesian model, $X_1, \dots, X_p$ are i.i.d. from $N(0, 1 + \tau^2)$. The UMVU estimate of $1 + \tau^2$ is $\sum_{i=1}^p X_i^2/p$, and slightly different multiples of $\|X\|^2$ may be sensible. The James–Stein estimator of θ is based on estimating $1/(1 + \tau^2)$ by $(p - 2)/\|X\|^2$ in (11.1). The resulting estimator is

$$\delta_{JS}(X) = \left(1 - \frac{p - 2}{\|X\|^2} \right) X. \quad (11.2)$$

The next section considers the risk of this estimator.

Although the derivation above has a Bayesian feel, the standard deviation τ that specifies the marginal prior distributions for the Θ_i is not modeled as a random variable. This deviation τ might be called a *hyperparameter*, and a fully Bayesian approach to this problem would treat τ as a random variable with its own prior distribution. Then given τ , $\Theta_1, \dots, \Theta_p$ would be conditionally i.i.d. from $N(0, \tau^2)$. This approach, specifying the prior by coupling a marginal prior distribution for hyperparameters with conditional distributions for the regular parameters, leads to hierarchical Bayes models. With modern computing, estimators based on these models can be practical and have gained popularity in recent years. Hierarchical models are considered in greater detail in Section 15.1.

11.2 Risk of the James–Stein Estimator¹

The following integration by parts identity is an important tool in our study of the risk of the James–Stein estimator. Fubini’s theorem provides a convenient way to establish an appropriate regularity condition for this identity.

Lemma 11.1 (Stein). *Suppose $x \sim N(\mu, \sigma^2)$, $h : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable (absolutely continuous is also sufficient), and*

$$E|h'(X)| < \infty. \quad (11.3)$$

Then

$$E(X - \mu)h(X) = \sigma^2 E h'(X).$$

Proof. Assume for now that $\mu = 0$ and $\sigma^2 = 1$. If the result holds for a function h it also holds for h plus a constant, and so we can assume without loss of generality that $h(0) = 0$. By Fubini’s theorem,

$$\begin{aligned} \int_0^\infty xh(x)e^{-x^2/2} dx &= \int_0^\infty x \left[\int_0^x h'(y) dy \right] e^{-x^2/2} dx \\ &= \int_0^\infty \int_0^\infty I\{y < x\} x h'(y) e^{-x^2/2} dy dx \\ &= \int_0^\infty h'(y) \left[\int_y^\infty x e^{-x^2/2} dx \right] dy \\ &= \int_0^\infty h'(y) e^{-y^2/2} dy. \end{aligned}$$

The regularity necessary in Fubini’s theorem to justify the interchange of the order of integration follows from (11.3). A similar calculation shows that

$$\int_{-\infty}^0 xh(x)e^{-x^2/2} dx = \int_{-\infty}^0 h'(y)e^{-y^2/2} dy.$$

Adding these together and dividing by $\sqrt{2\pi}$, $EXh(X) = Eh'(X)$ when $X \sim N(0, 1)$.

For the general case, let $Z = (X - \mu)/\sigma \sim N(0, 1)$. Then $X = \mu + \sigma Z$ and

$$\begin{aligned} E(X - \mu)h(X) &= \sigma EZh(\mu + \sigma Z) \\ &= \sigma^2 Eh'(\mu + \sigma Z) \\ &= \sigma^2 Eh'(X). \end{aligned} \quad \square$$

The next lemma generalizes the previous result to higher dimensions. If $h : \mathbb{R}^p \rightarrow \mathbb{R}^p$, let Dh denote the $p \times p$ matrix of partial derivatives,

¹ This section covers optional material not used in later chapters.

$$[Dh(x)]_{ij} = \frac{\partial h_i(x)}{\partial x_j}.$$

Also, let $\|Dh\|$ denote the Euclidean norm of this matrix,

$$\|Dh(x)\| = \left\{ \sum_{i,j} [Dh(x)]_{ij}^2 \right\}^{1/2}.$$

Lemma 11.2. *Let $X_1, \dots, X_p$ be independent with $X_i \sim N(\theta_i, 1)$, $i = 1, \dots, p$. If*

$$E\|Dh(X)\| < \infty,$$

then

$$E(X - \theta)'h(X) = E\text{tr}\{Dh(X)\}.$$

Proof. Using Stein's lemma (Lemma 11.1) and smoothing,

$$\begin{aligned} E(X_i - \theta_i)h_i(X) &= EE[(X_i - \theta_i)h_i(X) \mid X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_p] \\ &= EE\left[\frac{\partial h_i(X)}{\partial X_i} \mid X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_p\right] \\ &= E\frac{\partial h_i(X)}{\partial X_i} \\ &= E[Dh(X)]_{ii}. \end{aligned}$$

Summation over i gives the stated result. $\square$

The final result provides an unbiased estimator of the risk. Let $X_1, \dots, X_p$ be independent with $X_i \sim N(\theta_i, 1)$. Given an estimator $\delta(X)$ of θ , define $h(X)$ as $X - \delta(X)$ so that

$$\delta(X) = X - h(X). \quad (11.4)$$

For the James–Stein estimator (11.2),

$$h(X) = \frac{p-2}{\|X\|^2} X.$$

Theorem 11.3. *Suppose $X_1, \dots, X_p$ are independent with $X_i \sim N(\theta_i, 1)$ and that h and δ are related as in (11.4). Assume that h is differentiable and define*

$$\hat{R} = p + \|h(X)\|^2 - 2\text{tr}\{Dh(X)\}.$$

Then

$$R(\theta, \delta) = E_\theta \|\delta(X) - \theta\|^2 = E_\theta \hat{R},$$

provided $E_\theta \|Dh(X)\| < \infty$.

Proof. Using Lemma 11.2,

$$\begin{aligned}
 R(\theta, \delta) &= E_\theta \sum_{i=1}^p (X_i - \theta_i - h_i(X))^2 \\
 &= E_\theta \left[\sum_{i=1}^p (X_i - \theta_i)^2 + \sum_{i=1}^p h_i^2(X) - 2 \sum_{i=1}^p (X_i - \theta_i) h_i(X) \right] \\
 &= p + E_\theta \|h(X)\|^2 - 2E_\theta (X - \theta) \cdot h(X) \\
 &= p + E_\theta \|h(X)\|^2 - 2E_\theta \text{tr}\{Dh(X)\}. \quad \square
 \end{aligned}$$

For the James–Stein estimator (11.2),

$$h(X) = \frac{p-2}{\|X\|^2} X,$$

and so

$$h_i(x) = \frac{(p-2)x_i}{x_1^2 + \cdots + x_p^2}.$$

Since

$$\begin{aligned}
 \frac{\partial h_i(x)}{\partial x_i} &= \frac{p-2}{x_1^2 + \cdots + x_p^2} - \frac{(p-2)x_i(2x_i)}{(x_1^2 + \cdots + x_p^2)^2} \\
 &= \frac{p-2}{\|x\|^2} - \frac{2(p-2)x_i^2}{\|x\|^4}, \\
 \text{tr}\{Dh(x)\} &= \frac{p(p-2)}{\|x\|^2} - \frac{2(p-2) \sum_{i=1}^p x_i^2}{\|x\|^4} = \frac{(p-2)^2}{\|x\|^2}.
 \end{aligned}$$

Also,

$$\|h(x)\|^2 = \sum_{i=1}^p \frac{[(p-2)x_i]^2}{\|x\|^4} = \frac{(p-2)^2}{\|x\|^2}.$$

Thus, for the James–Stein estimator,

$$\hat{R} = p + \frac{(p-2)^2}{\|X\|^2} - 2 \frac{(p-2)^2}{\|X\|^2} = p - \frac{(p-2)^2}{\|X\|^2}. \quad (11.5)$$

By Theorem 11.3,

$$R(\theta, \delta_{JS}) = E_\theta \hat{R} = E_\theta \left[p - \frac{(p-2)^2}{\|X\|^2} \right] < p = R(\theta, X).$$

Hence when $p > 2$, the James–Stein estimator always has smaller compound risk than the estimator X . Because the risk function for δ_{JS} is better than the risk function for X , in the language of decision theory, developed in the next section, X is called inadmissible.

When $\|\theta\|$ is large, $\|X\|$ will be large with high probability. Then the James–Stein estimator and X will be very similar and will have similar risk. But when $\|\theta\|$ is small there can be a substantial decrease in risk using the James–Stein estimator instead of X . If $\theta = 0$, then

$$\|X\|^2 = \sum_{i=1}^p X_i^2 \sim \chi_p^2.$$

Integrating against the chi-square density, as in (4.10),

$$E_0 \frac{1}{\|X\|^2} = \frac{1}{p-2}.$$

Using this and (11.5),

$$R(0, \delta) = E_0 \left[p - \frac{(p-2)}{\|X\|^2} \right] = p - \frac{(p-2)^2}{p-2} = 2.$$

Regardless of the dimension of θ and X , at the origin $\theta = 0$, the James–Stein estimator has risk equal to two.

The results in this section can be extended in various ways. James and Stein (1961) derived the estimator (11.2) and also consider estimation when σ^2 is unknown. Extensions to ridge regression are reviewed in Draper and van Nostrand (1979). Stein’s identity in Lemma 11.1 can be developed for other families of distributions, and these identities have been used in various interesting ways. Chen (1975) and Stein (1986) use them to obtain Poisson limit theorems, and Woodroffe (1989) uses them for interval estimation and to approximate posterior distributions.

11.3 Decision Theory²

The calculations in the previous section show that X is inadmissible when the dimension p is three or higher, leaving open the natural question of what happens in one or two dimensions. In this section, several results from decision theory are presented and used to characterize admissible procedures and show that for compound estimation X is admissible when $p = 1$. A similar argument shows that X is also admissible when $p = 2$, although the necessary calculations in that case are quite delicate.

Formal decision theory begins with a parameter space Ω , an action space $\mathcal{A}$, a data space $\mathcal{X}$, a model $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$, and a loss function $L : \Omega \times \mathcal{A} \rightarrow [-\infty, \infty]$. For simplicity and convenience, we assume that $\mathcal{X} = \mathbb{R}^n$, that Ω and $\mathcal{A}$ are Borel subsets of Euclidean spaces, and that the loss function L is nonnegative and measurable, with $L(\theta, a)$ lower semicontinuous in a .

² This section covers optional material not used in later chapters.

A measurable function $\delta : \mathcal{X} \rightarrow \mathcal{A}$ is called a *nonrandomized decision rule*, and its *risk function* is defined as

$$R(\theta, \delta) = \int L(\theta, \delta(x)) dP_\theta(x) = E_\theta L(\theta, \delta(X)), \quad \theta \in \Omega.$$

The set of all nonrandomized decision rules is denoted $\mathcal{D}_n$.

A nonrandomized decision rule associates with each x an action $\delta(x)$. In contrast, a randomized decision rule associates with each x a probability distribution δ_x , the idea being that if $X = x$ is observed, a random action A will be drawn from δ_x . So $A|X = x \sim \delta_x$. Formally, δ should be a stochastic transition kernel, satisfying the regularity condition that $\delta_x(A)$ is a measurable function of x for any Borel set A . By smoothing, the risk function for δ can be defined as

$$\begin{aligned} R(\theta, \delta) &= E_\theta L(\theta, A) = E_\theta E_\theta [L(\theta, A)|X] \\ &= \iint L(\theta, a) d\delta_x(a) dP_\theta(x), \quad \theta \in \Omega. \end{aligned}$$

The set of all randomized decision rules is denoted $\mathcal{D}$.

Example 11.4 (Estimation). For estimating a univariate parameter $g(\theta)$ it is natural to take $\mathcal{A} = \mathbb{R}$ as the action space, and a decision rule δ would be called an estimator. Representative loss functions include squared error loss with $L(\theta, a) = [a - g(\theta)]^2$ and weighted squared error loss with $L(\theta, a) = w(\theta)[a - g(\theta)]^2$.

Example 11.5 (Testing). In testing problems, the action space is $\mathcal{A} = \{0, 1\}$, with action “0” associated with accepting $H_0 : \theta \in \Omega_0$ and action “1” associated with accepting $H_1 : \theta \in \Omega_1$. For each x , δ_x is a Bernoulli distribution, which can be specified by its “success” probability $\varphi(x) = \delta_x(\{1\})$. This provides a one-to-one correspondence between test functions φ and randomized decision rules. A representative loss function now might be zero-one loss in which there is unit loss for accepting the wrong hypothesis:

$$L(\theta, a) = I\{a = 1, \theta \in \Omega_0\} + I\{a = 0, \theta \in \Omega_1\}.$$

If the power function β is defined as

$$\beta(\theta) = P_\theta(A = 1) = E_\theta P_\theta(A = 1|X) = E_\theta \delta_X(\{1\}) = E_\theta \varphi(X),$$

then the risk function with this loss is

$$\begin{aligned} R(\theta, \delta) &= \begin{cases} P_\theta(A = 1), & \theta \in \Omega_0; \\ P_\theta(A = 0), & \theta \in \Omega_0, \end{cases} \\ &= \begin{cases} \beta(\theta), & \theta \in \Omega_0; \\ 1 - \beta(\theta), & \theta \in \Omega_1. \end{cases} \end{aligned}$$

A decision rule δ is called *inadmissible* if a competing rule δ^* has a better risk function, specifically if $R(\theta, \delta^*) \leq R(\theta, \delta)$ for all $\theta \in \Omega$ with $R(\theta, \delta^*) < R(\theta, \delta)$ for some $\theta \in \Omega$. If this happens we say that δ^* *dominates* δ . All other rules are called *admissible*. If minimizing risk is the sole concern, no one would ever want to use an inadmissible rule, and there has been considerable interest in characterizing admissible rules. Our first results below show that Bayes rules are typically admissible. More surprising perhaps are extensions, such as Theorem 11.8 below, showing that the remaining admissible rules are almost Bayes in a suitable sense. For notation, for a prior distribution Λ let

$$R(\Lambda, \delta) = \int R(\theta, \delta) d\Lambda(\theta), \quad (11.6)$$

the integrated risk of δ under Λ , and let

$$R(\Lambda) = \inf_{\delta \in \mathcal{D}} R(\Lambda, \delta), \quad (11.7)$$

the minimal integrated risk. Finally, the decision rule δ is called *Bayes* for a prior Λ if it minimizes the integrated risk, that is, if

$$R(\Lambda, \delta) = R(\Lambda). \quad (11.8)$$

At this stage it is worth noting that in definitions (11.6) and (11.7) the prior Λ does not really need to be a probability measure; the equations make sense as long as Λ is finite, or even if it is infinite but σ -finite. The definition of Bayes for Λ also makes sense for these Λ . But if the prior Λ is not specified, δ is called proper Bayes only if (11.8) holds for some probability distribution Λ . Of course, if Λ is finite and δ is Bayes for Λ it is also Bayes for the probability distribution $\Lambda(\cdot)/\Lambda(\Omega)$. Thus we are only disallowing rules that are Bayes with respect to an “improper” prior with $\Lambda(\Omega) = \infty$ in this designation.

The next two results address the admissibility of Bayes rules.

Theorem 11.6. *If a Bayes rule δ for Λ is essentially unique, then δ is admissible.*

Proof. Suppose $R(\theta, \delta^*) \leq R(\theta, \delta)$ for all $\theta \in \Omega$. Then, by (11.6), $R(\Lambda, \delta^*) \leq R(\Lambda, \delta)$, and δ^* must also be Bayes for Λ . But then, by the essential uniqueness, $\delta = \delta^*$, a.e. $\mathcal{P}$, and so $R(\theta, \delta) = R(\theta, \delta^*)$ for all $\theta \in \Omega$. $\square$

The next result refers to the *support* of the prior distribution Λ , defined as the smallest closed set B with $\Lambda(B) = 1$. Note that if the support of Λ is Ω and B is an open set with $\Lambda(B) = 0$, then B must be empty, since otherwise B^c would be a closed set smaller than Ω with $\Lambda(B^c) = 1$.

Theorem 11.7. *If risk functions for all decision rules are continuous in θ , if δ is Bayes for Λ and has finite integrated risk $R(\Lambda, \delta) < \infty$, and if the support of Λ is the whole parameter space Ω , then δ is admissible.*

Proof. Suppose again that $R(\theta, \delta^*) \leq R(\theta, \delta)$ for all $\theta \in \Omega$. Then, as before, δ^* is Bayes for Λ and δ and δ^* must have the same integrated risk, $R(\Lambda, \delta) = R(\Lambda, \delta^*)$. Hence

$$\int (R(\theta, \delta) - R(\theta, \delta^*)) d\Lambda(\theta) = 0.$$

Since the integrand here is nonnegative, by integration fact 2 in Section 1.4 the set

$$\{\theta : R(\theta, \delta) - R(\theta, \delta^*) > 0\}$$

has Λ measure zero. But since risk functions are continuous, this set is open and must then be empty since Λ has support Ω . So the risk functions for δ and δ^* must be the same, $R(\theta, \delta) = R(\theta, \delta^*)$ for all $\theta \in \Omega$. $\square$

A collection of decision rules is called a *complete class* if all rules outside the class are inadmissible. A complete class will then contain all of the admissible rules. In various situations suitable limits of Bayes procedures form a complete class. Because randomized decision rules are formally stochastic transition functions, a proper statement of most of these results involves notions of convergence for these objects, akin to our notion of convergence in distribution for probability distributions, but complicated by the functional dependence on X . An exception arises if the loss function $L(\theta, a)$ is strictly convex in a . In this case, admissible rules must be nonrandomized by the Rao–Blackwell theorem (Theorem 3.28), and we have the following result, which can be stated without reference to complicated notions of convergence. This result appears with a careful proof as Theorem 4A.12 of Brown (1986). Let $\mathcal{B}_0$ denote the class of Bayes rules for priors Λ concentrated on *finite* subsets of Ω .

Theorem 11.8. *Let $\mathcal{P}$ be a dominated family of distributions with p_θ as the density for P_θ , and assume that $p_\theta(x) > 0$ for all $x \in \mathcal{X}$ and all $\theta \in \Omega$. If $L(\theta, \cdot)$ is nonnegative and strictly convex for all $\theta \in \Omega$, and if $L(\theta, a) \rightarrow \infty$ as $\|a\| \rightarrow \infty$, again for all $\theta \in \Omega$, then the set of pointwise limits of rules in $\mathcal{B}_0$ forms a complete class.*

This and similar results show that in regular cases any admissible rule will be a limit of Bayes rules. Unfortunately, some limits may give inadmissible rules, and these results cannot be used to show that a given rule is admissible. The final theoretical result of this section gives a sufficient condition for admissibility. For regularity, it assumes that all risk functions are continuous, but similar results are available in different situations. Let

$$\overline{B}_r(x) = \{y : \|y - x\| \leq r\},$$

the closed ball of radius r about x .

Theorem 11.9. *Assume that risk functions for all decision rules are continuous in θ . Suppose that for any closed ball $\overline{B}_r(x)$ there exist finite measures Λ_m such that $R(\Lambda_m, \delta) < \infty$, $m \geq 1$,*

$$\liminf A_m(\overline{B}_r(x)) > 0,$$

and

$$R(\delta, A_m) - R(A_m) \rightarrow 0.$$

Then δ is admissible.

Proof. Suppose δ^* dominates δ . Then $R(\theta_0, \delta^*) < R(\theta_0, \delta)$ for some $\theta_0 \in \Omega$. By continuity

$$\inf_{\theta \in \overline{B}_r(\theta_0)} [R(\theta, \delta) - R(\theta, \delta^*)] \rightarrow R(\theta_0, \delta) - R(\theta_0, \delta^*) > 0$$

as $r \downarrow 0$, and so there exist values $\epsilon > 0$ and $r_0 > 0$ such that

$$R(\theta, \delta) \geq R(\theta, \delta^*) + \epsilon, \quad \forall \theta \in \overline{B}_{r_0}(\theta_0).$$

Since δ^* dominates δ , this implies

$$R(\theta, \delta) \geq R(\theta, \delta^*) + \epsilon I\{\theta \in \overline{B}_{r_0}\}.$$

Integrating this against A_m ,

$$R(A_m, \delta) \geq R(A_m, \delta^*) + \epsilon A_m(\overline{B}_{r_0}) \geq R(A_m) + \epsilon A_m(\overline{B}_{r_0}),$$

contradicting the assumptions of the theorem. $\square$

Stein (1955) gives a necessary and sufficient condition for admissibility, and using this result the condition in this theorem is also necessary. Related results are given in Blyth (1951), Le Cam (1955), Farrell (1964, 1968a,b), Brown (1971b), and Chapter 8 of Berger (1985).

Example 11.10. Consider a Bayesian formulation of the one-sample problem in which $\Theta \sim N(0, \tau^2)$ and given $\Theta = \theta$, $X_1, \dots, X_n$ are i.i.d. from $N(\theta, \sigma^2)$ with σ^2 a known constant. By the calculation for Problem 6.21, the posterior distribution for Θ is

$$\Theta|X = x \sim N\left(\frac{\overline{x}}{1 + \sigma^2/(n\tau^2)}, \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}\right),$$

where $\overline{x} = (x_1 + \dots + x_n)/n$. So the Bayes estimator under squared error loss is

$$\frac{\overline{X}}{1 + \sigma^2/(n\tau^2)}$$

with integrated risk

$$\frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}.$$

Since the Bayes estimator converges to $\overline{X}$ as $\tau \rightarrow \infty$, if we are hoping to use Theorem 11.9 to show that the sample average $\delta = \overline{X} = (X_1 + \dots + X_n)/n$

is admissible, it may seem natural to take $\Lambda_m = N(0, m)$. But this does not quite work; since densities for these distributions tend to zero, with this choice $\Lambda_m(\overline{B}_r(x))$ tends to zero as $m \rightarrow \infty$. The problem can be simply fixed by rescaling, taking $\Lambda_m = \sqrt{m}N(0, m)$. The density for this measure is $\phi(\theta/\sqrt{m})$, converging pointwise to $\phi(0) = 1/\sqrt{2\pi}$. So by dominated convergence,

$$\Lambda_m(\overline{B}_r(x)) = \int_{x-r}^{x+r} \phi(\theta/\sqrt{m}) d\theta \rightarrow \frac{2r}{\sqrt{2\pi}}.$$

Scaling the prior by $\sqrt{m}$ scales risks and expectations by the same factor $\sqrt{m}$, and so

$$R(\delta, \Lambda_m) = \frac{\sqrt{m}\sigma^2}{n} \quad \text{and} \quad R(\Lambda_m) = \sqrt{m}\sigma^2 \frac{m}{\sigma^2 + nm}.$$

Then

$$R(\delta, \Lambda_m) - R(\Lambda_m) = \frac{\sqrt{m}\sigma^4}{n(\sigma^2 + nm)} \rightarrow 0$$

as $m \rightarrow \infty$, and by Theorem 11.9 $\overline{X}$ is admissible.

Stein (1956) shows admissibility of the sample average $\overline{X}$ in $p = 2$ dimensions. The basic approach is similar to that pursued in this example, but the priors Λ_n must be chosen with great care; it is not hard to see that scaled conjugate normal distributions will not work.

For a more complete introduction to decision theory, see Chernoff and Moses (1986) or Bickel and Doksum (2007), and for a more substantial treatment, see Berger (1985), Ferguson (1967), or Miescke and Liese (2008).

11.4 Problems³

- *1. Consider estimating the failure rates $\lambda_1, \dots, \lambda_p$ for independent exponential variables $X_1, \dots, X_p$. So X_i has density $\lambda_i e^{-\lambda_i x}$, $x > 0$.
 - a) Following a Bayesian approach, suppose the unknown parameters are modeled as random variables $\Lambda_1, \dots, \Lambda_p$. For a prior distribution, assume these variables are i.i.d. from a gamma distribution with shape parameter α and unit scale parameter, so Λ_i has density $\lambda^{\alpha-1} e^{-\lambda} / \Gamma(\alpha)$, $\lambda > 0$. Determine the marginal density of X_i in this model.
 - b) Find the Bayes estimate of Λ_i in the Bayesian model with squared error loss.
 - c) The Bayesian model gives a family of joint distributions for $X_1, \dots, X_p$ indexed solely by the parameter α (the joint distribution does not depend on $\lambda_1, \dots, \lambda_p$). Determine the maximum likelihood estimate of α for this family.

³ Solutions to the starred problems are given at the back of the book.

- d) Give an empirical Bayes estimator for λ_i combining the “empirical” estimate for α in part (c) with the Bayes estimate for λ_i when α is known in part (b).
- *2. Consider estimation of regression slopes $\theta_1, \dots, \theta_p$ for p pairs of observations, $(X_1, Y_1), \dots, (X_p, Y_p)$, modeled as independent with $X_i \sim N(0, 1)$ and $Y_i|X_i = x \sim N(\theta_i x, 1)$.
- Following a Bayesian approach, let the unknown parameters $\Theta_1, \dots, \Theta_p$ be i.i.d. random variables from $N(0, \tau^2)$. Find the Bayes estimate of Θ_i in this Bayesian model with squared error loss.
 - Determine EY_i^2 in the Bayesian model. Using this, suggest a simple method of moments estimator for τ^2 .
 - Give an empirical Bayes estimator for θ_i combining the simple “empirical” estimate for τ in (b) with the Bayes estimate for θ_i when τ is known in (a).
- *3. Consider estimation of the means $\theta_1, \dots, \theta_p$ of p independent Poisson random variables $X_1, \dots, X_p$ under compound squared error loss, $L(\theta, d) = \sum_{i=1}^p (\theta_i - d_i)^2$.
- Following a Bayesian approach, let the unknown parameters be modeled as random variables $\Theta_1, \dots, \Theta_p$ that are i.i.d. with common density $\lambda e^{-\lambda x}$ for $x > 0$, 0 for $x \leq 0$. Determine the Bayes estimators of $\Theta_1, \dots, \Theta_p$.
 - Determine the marginal density (mass function) of X_i in the Bayesian model.
 - In the Bayesian model, $X_1, \dots, X_p$ are i.i.d. with the common density in part (b). Viewing this joint distribution as a family of distributions parameterized by λ , what is the maximum likelihood estimator of λ .
 - Suggest empirical Bayes estimators for $\theta_1, \dots, \theta_p$ based on the Bayesian estimators in part (a) with an empirical estimator of λ from part (c).
4. Consider estimating success probabilities $\theta_1, \dots, \theta_p$ for p independent binomial variables $X_1, \dots, X_p$, each based on m trials, under compound squared error loss, $L(\theta, d) = \sum_{i=1}^p (\theta_i - d_i)^2$.
- Following a Bayesian approach, model the unknown parameters as random variables $\Theta_1, \dots, \Theta_p$ that are i.i.d. from a beta distribution with parameters α and β . Determine the Bayes estimators of $\Theta_1, \dots, \Theta_p$.
 - In the Bayesian model, $X_1, \dots, X_p$ are i.i.d. Determine the first two moments for their common marginal distribution, EX_i and EX_i^2 . Using these, suggest simple method of moments estimators for α and β .
 - Give empirical Bayes estimators for θ_i combining the simple “empirical” estimates for α and β in (b) with the Bayes estimate for θ_i when α and β are known in (a).
5. Consider estimation of unknown parameters $\theta_1, \dots, \theta_p$ based on data $X_1, \dots, X_p$ that are independent with $X_i \sim N(\theta_i, 1)$ under compound squared error loss.

- a) Following a Bayesian approach, model the unknown parameters as random variables $\Theta_1, \dots, \Theta_p$ that are i.i.d. from $N(\nu, \tau^2)$. Find Bayes estimators for the random parameters $\Theta_1, \dots, \Theta_p$.
 - b) Suggest “empirical” estimates for ν and τ^2 based on $\bar{X}$ and S^2 , the mean and sample variance of the X_i .
 - c) Give empirical Bayes estimators for $\theta_1, \dots, \theta_p$ based on the Bayesian estimators in (a) and the estimates for ν and τ^2 in (b).
6. Consider estimation of unknown parameters $\theta_1, \dots, \theta_p$ based on data $X_1, \dots, X_p$ that are independent with $X_i \sim \text{Unif}(0, \theta_i)$, $i = 1, \dots, p$, under compound squared error loss.
- a) Following a Bayesian approach, model the unknown parameters as random variables $\Theta_1, \dots, \Theta_p$ which are i.i.d. and absolutely continuous with common density

$$x\lambda^2 1_{(0,\infty)}(x)e^{-\lambda x}.$$

Find Bayes estimators for $\Theta_1, \dots, \Theta_p$.

- b) Suggest an empirical estimate for λ based on the sample average $\bar{X}$.
- c) Give empirical Bayes estimators for $\theta_1, \dots, \theta_p$ based on the Bayes estimators in (a) and the estimator for λ in (b).